OM BHATT

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PROFESSIONAL SUMMARY

Well-rounded student with strong academics and extracurricular experience. Passionate about robotics, particularly autonomous and intelligent systems. Seeking to expand engineering knowledge and experience to design, develop, and optimize mechatronic and robotic systems that advance automation and contribute to the betterment of society.

U.S. Person (No Sponsorship Required)

EDUCATION

Texas A&M University, *B.S - Mechatronics Engineering* — 2024 - 2028
Minor: Embedded Systems — *College Station, TX*

Relevant Coursework: Calculus I–III, Differential Equations, Electricity & Magnetism, Engineering Mechanics, Strength of Materials, Metallic Materials, Fluid Mechanics & Power, Circuit Analysis, Analog Electronics, Digital Electronics, Embedded Systems Development in C, Microcontroller Architecture, Applied Dynamic Systems

Certifications: Introduction to Programming using Python

Organizations: IEEE TAMU, ASME TAMU, TURTLE Robotics, Texas Aimbots, SOMTECH, TAMUHACK, SASE

RELEVANT EXPERIENCE

R&D Robotics Engineer - Software Subteam — Sep 2025 - Present
T.U.R.T.L.E Robotics, Disaster Response Observation Network, Texas A&M University — *College Station, TX*

- Researched and programmed autonomous flight algorithms (offboard_control_square_2) using PX4/MAVROS over ROS 2 Humble, validating autonomous flight via 2-drone simulation and 2-3 outdoor test flights for a 10-member team's UAV swarm platform assisting first responders.
- Co-authored and presented the "Disaster Response Observation Network (DRON)" research poster across two university showcases (Nov 2025, Apr 2026), each featuring 21+ advanced projects, detailing autonomous swarm progress and system architecture to an audience of engineering faculty and peers.

Hardware Team Member — Sep 2025 - Present
Texas Aimbots, Texas A&M University — *College Station, TX*

- The Sentry Robot is a fully autonomous ground robot that moves and shoots at enemy robots using computer vision.
- Designed and 3D-printed 4+ ABS structural components in Onshape CAD for Sentry, a fully autonomous ground robot, including a gimbal housing redesigned for a more compact, rigid footprint; installed on the robot.
- Integrated the NVIDIA Jetson Orin, LiDAR, and camera systems to enable real-time perception and autonomous decision-making ahead of the 2026 ARC Championship, an international collegiate robotics competition.

AEOP Summer Robotics Research Intern - LLMs for Safe Navigation REU — June 2025 - Aug 2025
Oden Institute - Center for Autonomy - Autonomous Systems Group, UT Austin — *Austin, TX*

- Researched autonomous off-road navigation and drift/jump control dynamics under Dr. Ufuk Topcu, Dr. Christian Ellis, and PhD Student Rwik Rana to architect a low-cost, 1/16 scale autonomous RC car as a modular research platform for drift and jump navigation studies.
- Executed the complete electromechanical assembly by designing and soldering a custom circuit board to power and integrate all components, including a Raspberry Pi 5/Jetson Nano and motor controllers, verified via multimeter testing.

- Programmed a low-level C++/ROS 2 controller for Ackermann steering and throttle control, validating full teleoperation across 5+ test runs, including a successful jump test.
- Ran a visual SLAM algorithm using camera-based perception in the F1Tenth simulator to validate localization and control, laying the groundwork for planned IMU fusion and post-jump re-localization toward fully autonomous operation.

Hatchling Team Member Jan 2025 - Sep 2025
T.U.R.T.L.E Robotics, Texas A&M University *College Station, TX*

- Won 2nd Place out of 30+ competing teams at the 2025 TURTLE Robotics Hatchling Competition
- Designed and CADded key mechanical components in SolidWorks, including a forklift-style lift and single-block gripper mechanism, as part of a 4-person team, engineered for rapid sequential block acquisition.
- Programmed an ESP32 and Arduino to control precise vehicle navigation and operate the multi-stage lift, enabling the robot to retrieve and stack 7 blocks in competition.
- Collaborating with the T.U.R.T.L.E Robotics team at Texas A&M University to enhance robotics expertise and tackle innovative challenges.

Student Researcher: Engineering Science Fairs - Dual-axis Solar Tracker Sep 2022 - Mar 2024
Science Fair Club, Vista Ridge High School *Cedar Park, TX*

- Engineered and developed "EcoTrack", a novel phototropic solar tracker inspired by nature, which uses thermo-responsive polymers to follow a light source. This design eliminates the need for electrical components in the tracking mechanism, creating a low-cost and low-maintenance solution for developing nations.
- Validated the system through experimentation, demonstrating that EcoTrack harnessed 87 percent more net energy than a traditional dual-axis tracker and 119 percent more than a fixed panel system under identical conditions.
- Achieved first place across Austin region at Austin Regional Science Fair and advanced to TXSEF State Fair
- Awarded a university tuition scholarship from Jacobs Engineering ($2000) and RICOH Sustainable Development Award
- Elected to lead the role of an officer to help mentor other students on their projects and showcase our school's achievements to attract potential sponsors

Mechatronics Research Intern - WATonoBus: Autonomous Shuttle Jun 2021 - Sep 2021
Mechatronics Vehicle Systems Lab, University of Waterloo *Waterloo, Canada*

- Engineered an external HMI (eHMI): an LED matrix display for WATonoBus, Professor Amir Khajepour's autonomous shuttle at MVS Lab, as 1 of 4 team members (2 students, 1 PhD student) on the eHMI subsystem.
- Developed a ROS node parsing intention-based commands over a rosserial link, validating all icon states and display functions in a live outdoor test on the shuttle's campus loop route.

PUBLICATIONS

- I. Wilhite, C. Ambroziak, A. Briggs, R. Kato, **O. Bhatt**, L. Breedlove, V. So, R. Shah, M. Shi, J. Shouba, O. Thomas, C. Weatherspoon, and D. Wheaton, "Disaster Response Observation Network (DRON)," TURTLE Robotics, Apr. 2026. [Conference Poster].
- I. Wilhite, A. Briggs, C. Ambroziak, R. Kato, D. Wheaton, **O. Bhatt**, A. Talal, J. Shouba, C. Xu, O. Thomas, and S. Ahn, "Disaster Response Observation Network (DRON)," TURTLE Robotics, Nov. 2025. [Conference Poster].
- **O. Bhatt**, "Autonomous RC Rally Car for Jump and Drift Racing," under the guidance of Dr. U. Topcu, Dr. C. Ellis, and R. Rana, AEOP Summer Robotics Research REU, UT Austin, Aug. 2025. [Research Presentation & Abstract]

VOLUNTEERING AND LEADERSHIP

Autodesk Ambassador Program Jan 2026 - Present
Autodesk, Texas A&M University *College Station, TX*

- Selected for a competitive ambassador program to represent Autodesk, promoting CAD tools through campus events and workshops.

Design Review Officer Jan 2026 - Present
T.U.R.T.L.E Robotics, Project Branch, Texas A&M University *College Station, TX*

- Evaluate system requirements, CAD/electronics documentation, and BOMs across 20+ projects over 3 Design Review sessions.

Sponsorship Committee Member Sept 2025 - Jan 2026
T.U.R.T.L.E Robotics, External Branch, Texas A&M University *College Station, TX*

- Securing corporate sponsorships for TURTLE, a student-led organization, through a new Corporate Sponsor Outreach Initiative.
- Developing and executing outreach strategies to engage with companies and secure funding for workshops, social events, and advanced technical projects.

Texas Science and Engineering Fair Judge Mar 2025 - Present
College of Engineering, Texas A&M University *College Station, TX*

- Evaluate 20+ student research projects annually in the Energy: Sustainable Materials & Designs (2025) and Robotics & Intelligent Machines (2026) categories at the Texas Science and Engineering Fair TXSEF, providing constructive feedback to the next generation of engineers.
- As part of Judge appreciation, George H.W. Bush Combat Development Complex gave TXSEF 2025 judges a tour of the Research Integration Center, which has state-of-the-art technology.

Student Council Member Sep 2022 - Sep 2024
Student Council Club, Vista Ridge High School *Cedar Park, TX*

- Contributed over 20 hours of community service by participating and helping organize events including school Christmas decorations, district events, conducting fundraisers, elementary school outreach events and more
- Donated items such as toys and food for charities

SKILLS

Robotics & Autonomy: ROS 2, PX4, MAVROS, SLAM, OpenCV, PWM/PID Control, Teleoperation, Gazebo, RViz, Point Clouds

Hardware and Prototyping: ESP32, Arduino, Raspberry Pi, STM32, FPGA, LiDAR, Cameras, PCB Design, Soldering, Analog Electronics, Oscilloscope & DMM Testing, Data Acquisition Systems

Programming Languages: Python, C/C++, Embedded C, Linux, Shell (Bash/Zsh), LaTeX, HTML, CSS

Simulation and CAD: SolidWorks, FEA, Autodesk Inventor, Onshape, Bambu Studio, Eagle, Multisim, MATLAB/Simulink

AI & Software Tools: Git/GitHub, Visual Studio Code, Workspace, Microsoft Office, LLM Integration (ChatGPT, Claude, Gemini)

AWARDS

AEOP Summer Robotics Research REU Scholarship ($6000), Oden Institute Jul 2025
Presented by Army Educational Outreach Program (AEOP) and Oden Institute at the University of Texas at Austin for my summer REU work.

2nd Place - 2025 TURTLE Robotics Hatchling Competition, T.U.R.T.L.E Robotics — May 2025

Won 2nd Place at the 2025 TURTLE Robotics Hatchling Competition.

AP Scholar, College Board — May 2024

Granted to students who receive scores of 3 or higher on three or more AP Exams.

Vista Ridge High School Scholar: "A" Honor Roll, Vista Ridge High School — May 2024

Presented to students who have straight As at the end of the school year in all classes enrolled

Texas Science & Engineering Fair Finalist, TXSEF — Mar 2024

Advanced to the Texas Science & Engineering Fair (Statewide Science Fair)

2nd Place at Austin Regional Science Fair, Austin Energy — Feb 2024

Placed second in the Energy: Sustainable Materials and Design category across the Austin metropolitan region

Vista Ridge High School Scholar: "A" Honor Roll, Vista Ridge High School — May 2023

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Jacobs Engineering Tuition Scholarship ($2000), Jacobs Engineering — Feb 2023

Presented to one selected finalist as a tuition scholarship that can be applied to the college of choice

RICOH Sustainable Development Award, RICOH — Feb 2023

Presented to the student whose research has demonstrated the principles and technical innovations that offer the greatest potential for sustainable development

Summer Intern Scholarship, University of Waterloo — Jun 2021

Presented by Professor Amir Khajepour to me for my work at Mechatronics Vehicle Systems Lab